1 A useful identity

The sum in Equation 1 follows by pairing the first and last terms.

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}. \tag{1}$$

$$(a+b)^2 = a^2 + 2ab + b^2, \tag{2}$$

$$\int_0^1 x^2\,dx = \frac{1}{3}. \tag{3}$$